Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/17/2004
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Project Title:	Direct Visualization and Light Scattering Studies of the Structure, Dynamics and Rheology of Anisometric Colloidal Suspensions and Gels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2136	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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Task Description:	This project studies the aggregation and assembly of anisometric colloidal suspensions by confocal microscopy and light scattering. Understanding of anisometric colloid structure is useful for the development of new materials with anisotropic optical, permeability and mechanical properties for applications such as filtration, purification and sensing.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	1. Science Progress and Status Confocal microscopy of colloidal rod suspensions. Stable suspensions of monodisperse fluorescent polymeric microspheroids have been synthesized as model systems for the direct visualization of amorphous and ordered structures in suspensions of anisometric colloids. The method is adapted from Keville et al and involves the preparation of monodisperse PMMA microspheres, which are subsequently deformed in a uniaxial extensional flow to generate anisometric particles – ellipsoids and spherocylinders. The size of these anisometric particles (a~ 5 um, b ~ 0.8 um) makes them ideal for confocal laser scanning micrscopy (CLSM) studies. They can be dispersed in refractive-index matching mixed organic solvents to minimize the effects of scattering and thereby allow imaging deep into the sample volume. Depletion interactions between the particles can be readily induced in these organic suspensions by the addition of a non-adsorbing polymer. An image processing code for the detection of the centroid and orientation angle of rod particles in a three-dimensional image volume (as acquired by confocal microscopy) has been completed. This method yields totally new information about orientation angle distributions in rod suspensions, gels and sediments. We have tested the image processing code against rod behavior in a stretched film and oscillatory flow. Interestingly, we found nematic-like ordering in sedimented suspensions of rods with aspect ratio 5.4. The nematic-like ordering opens exciting possibilities for anisotropic optical and mechanical properties in colloidal materials that are analogous to those seen for molecular liquid crystals. Because of their anisotropic ordering, these materials are of potential use as sensors. We developed a method to quantify the long-range structure of colloidal dispersions from confocal microscopy by measuring the low-wave vector limit of the static structure factor. This measure characterizes the compressibility of the gel structure. The system used to test the method is comprised of PMMA-g-PDMS microspheres stably dispersed in mixtures of the solvents cyclohexyl bromide and decalin. Polystyrene is also introduced as a depletion agent to generate an attractive pair potential. (The particles were synthesized using the method as prescribed by Keville et al.) Having verified the method for both gel and crystalline samples, it is being applied to studies of anisometric and spherical colloidal gels. A flow cell is being used to study the effect of shear stress on structure and kinetics of gels of high colloidal volume fraction spherical particle gels (colloid volume fraction = 0.1). A low shear rate (10 microns/sec) is applied over a gap of about 250 microns. Confocal microscopy is used to acquire 2D images that quantify the change in gel structure during and after shear flow. The system under investigation is PMMA-g-PDMS microspheres stably dispersed in mixtures of the solvents cyclohexyl bromide and decalin. Polystyrene is used as depletion agent. Light scattering studies of colloidal rod suspensions. We synthesized model, monodisperse boehmite colloidal rods of aspect ratio between 4-30 and showed by light scattering that the fractal dimension for their diffusion-limited aggregation displays an unexpected dependence on rod aspect ratio. The result was corroborated by Monte Carlo simulation and an explanation in terms of rod anisotropic volume was developed. This research was reported in Physical Review Letters. A second publication is in preparation. We have also conducted experiments to quantify the transient structural evolution of colloidal DLCA clusters during start-up of steady shear flow. This work identifies previously unforeseen effects of the nature of inter-particle interactions on the non-linear rheology of fractal colloidal gels. It makes use of a previously built Taylor-Couette shear cell and ultra-small angle light scattering devices. This publication is under review at Journal of Rheology. 2. Plans for 2005 Our work will focus on three areas: (1) Order/disorder transitions in rod suspensions. We will generate a phase diagram that better quantifes the bounds of the region of the order/disorder orientation transition (nematic-like) that we have discovered in fluorescent suspensions of colloidal rods. Although we have found ordering under a few conditions, the subtle trade off between ordering and jamming needs to be better understood if these materials are to be developed for their anisotropic optical properties: (2) Depletion gelation of rods. To study the rheology of systems comprising of anisometric particles, we will engage in the preparation of gels of rod shaped particles using spheres as the depletion agent. We are using multi-channel fluorescence to simultaneously and independently image the rods and sphere by confocal microscopy; (3) Rheology of anisometric gels. We will apply soft glassy rheology models to understand the unusual transient rheology of gels comprised of non-spherical particles. We will combine this study with measurements of aging in gels.
Bibliography Type:	Description: (Last Updated: 10/29/2019)
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Presentation	Solomon MJ "Confocal microscopy, light scattering and particle shape effects in gels of colloidal rods" Levich Insitute City College of New York Oct-2004
Presentation	Solomon MJ "Flow-assisted assembly of ordered colloidal structures" Department of Chemical Engineering, Michigan Tech University Oct-2004